

To-do

- Generate two synthetic samples WT and KO with $N = 1000$ observations each.
- define two self made statistics
- test with permutation tests
- estimate their confidence intervals with bootstrap sampling

Step 1 Generate observations

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import skew

np.random.seed(0) #same random output each time it runs

#step 1 : generate WT and KO samples
# Wt is a plain standard normal and KO is mostly standard normaly but a
fraction of its values are swapped for outcomes from a shifted high response
distribution N(3,0.5)

def make_samples(N,p_high):
    WT = np.random.normal(0,1,N)
    n_high = int(round(N*p_high)) #how many KO are high responders
    n_base = N - n_high #rest = ordinary
    KO_base = np.random.normal(0,1,n_base)
    KO_high = np.random.normal(3, 0.5 , n_high)
    KO = np.concatenate([KO_base, KO_high]) np.random.shuffle(KO) #so the
high responders aren't just sitting at the end of the array return WT, KO
```

Discussion

Why does this data-generating process create a difference in the upper tail of the distribution?

The KO group is a mix of two distributions ie 90 percent of it is WT ($N(0,1)$) and 10 percent is swapped for a distribution that is centered higher ie $(3,0.5)$. Since only the high response subset shifts, it lives above the rest in the form of extra density in the upper tail. The bulk of the KO distribution (the 90%) is still like WT.

Would this difference necessarily be well described by a change in the mean?

No, only 10 percent of KO is shifted, overall mean shifts slightly while the shape of the distribution changes in the tail. A mean based test could miss or understate this since mean tests would treat a unique subpopulation the same even all points were to be shifted by a little.

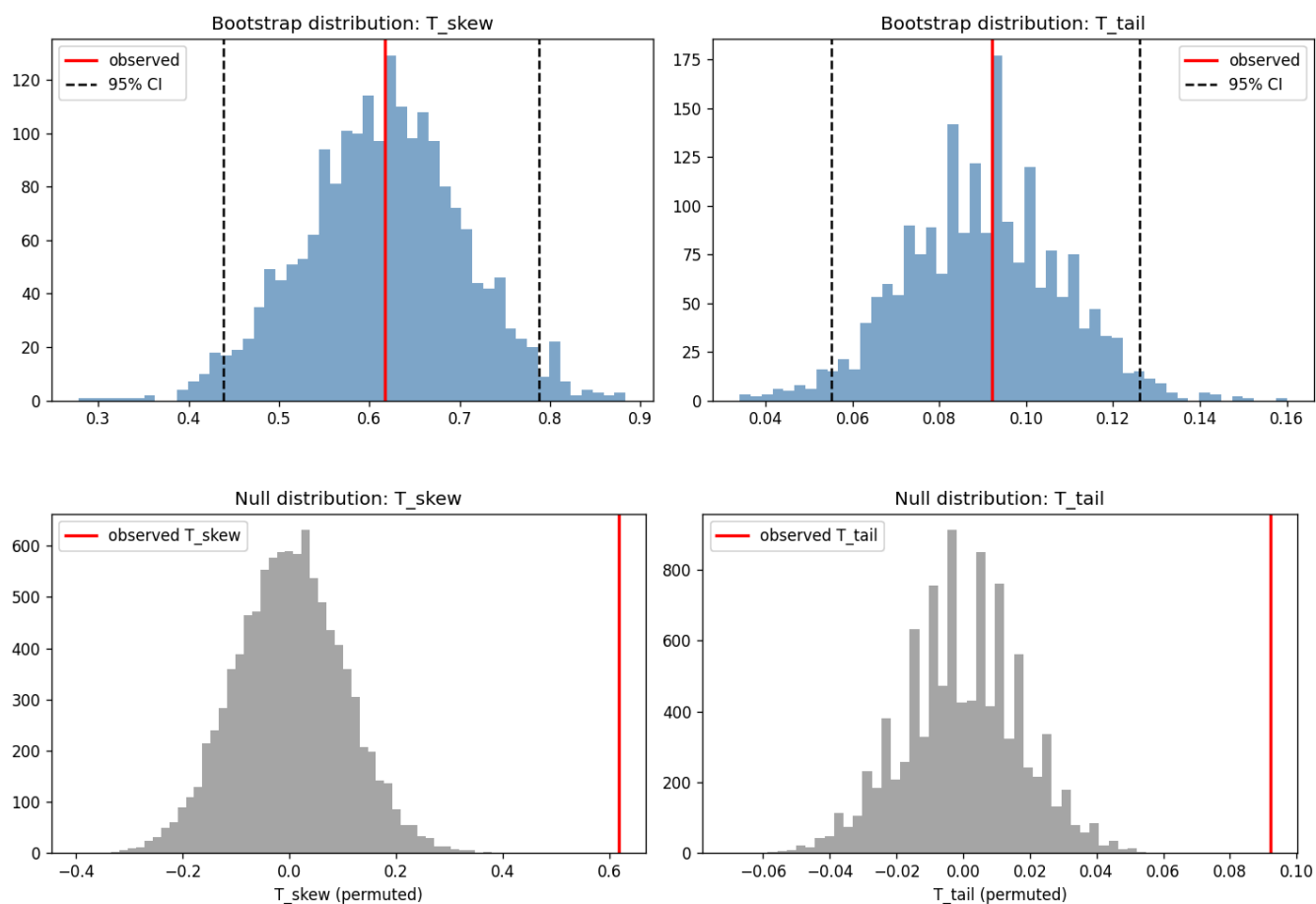
```
#two self made statistics

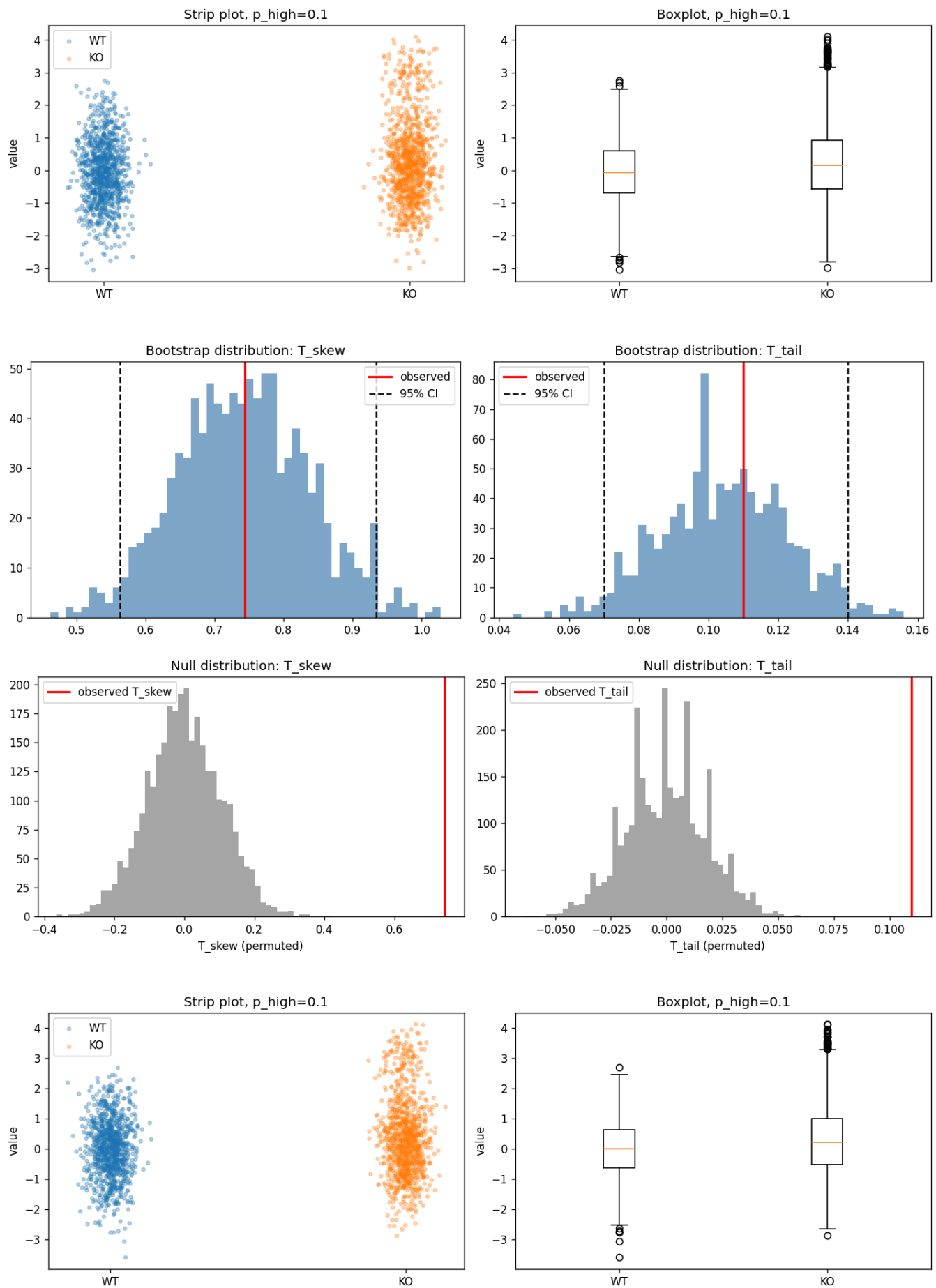
def T_SkStat(WT , KO):
    return skew(KO)-skew(WT)

def T_tail_stat(WT, KO):
    pooled = np.concatenate( [WT,KO])
    q80 = np.percentile(pooled, 80)
    frac_KO = np.mean(KO > q80)
    frac_WT = np.mean(WT > q80)
    return frac_KO - frac_WT
```

Visualise your samples

(scatter plot + box plot)





Discussion

Do WT and KO differ across the whole distribution, or mainly in the upper tail?

Mainly in the upper tail, in the scatter plot main cloud of KO points sit at the same level as WT except for the few KO points that extend between 3 and 4 that WT does not show

Is the KO distribution symmetric?

The boxplot is a clear answer to this question. They are not symmetric

Can you visually detect the high-response subpopulation?

Yes, in the scatter plot, the cluster of orange points sitting apart from the main KO cloud below (3-4). This is not a feature of the box plot visualization however.

Step 3: First statistic - difference in skewness

Discussion

What biological question does this represent?

If KO has an asymmetric right skewed shape compared to WT = if a subset of cells or synapses in KO respond rather strongly than the entire population shifting together

Why skewness might be more informative than the mean?

The mean averages unique subgroup effects across the entire sample and therefore dilutes it. Skewness is an asymmetry test concurrent with how a unique causal population could behave.

What would make the T skew large and positive?

A KO db with a very apparent right tail and has few values that are extremely high while WT stays symmetric.

Step 4: Second statistic: upper-tail enrichment

Discussion

What biological question does this represent?

are KO cells over-represented among the high responders in the top 20%

Why might this statistic be useful for detecting a small subpopulation of strongly responding cells?

It directly counts what happens in the exact region where the rare subpopulation lives, rather than interpreting it indirectly from an overall shape summary.

Why is this statistic not equivalent to comparing group means?

It only looks at what happens above a fixed cutoff and ignores the rest of the distribution. Also, Two groups can have identical means but very different T_{tail} value

Step 5: Permutation Test

Discussion

Is the observed statistic near the centre of the null distribution or in one of its tails?

Both observed values (the red lines) sit outside the whole null distribution, far off in the right tail.

Do both statistics lead to the same conclusion?

Both indicate KO differs from WT in the same direction

Which statistic seems better matched to the way the data were generated?

T_{tail} is the better option for the way the data was generated because the data-gen process inserts points into the upper tail and t_{tail} counts extra points there consequently. The t_{skew} captures the same phenomenon but with a shape summary.

Step 6: Permutation p-values

Discussion

Which statistic gives the smaller p-value?

when p_{high} is 0.10 both hit the floor of the everything that 10000 permutations can measure (0/10000). They only separate at smaller effect sizes

Why can two different self-made statistics give different p-values for the same data?

They're sensitive to different qualities of the group difference, so the same dataset can look relatively extreme depending on which lens measures it.

Which p-value would you report, and why?

T_{tail} . Because t 's more directly interpretable and answers the actual biological question being asked effectively

Step 7: Bootstrap Confidence Intervals

Discussion

Is zero inside the bootstrap confidence interval?

No

Are the confidence intervals narrow or wide?

Both are moderately narrow relative to the observed value, approximately half of the observed magnitude in width. excluding zero and there is some room left

Do the bootstrap confidence intervals agree with the permutation p-values?

Yes, both p-values were 0, and both Confidence intervals exclude zero, two ways of showing the same signal

What is the conceptual difference between a permutation null distribution and a bootstrap sampling distribution?

The permutation null answers the question: if group membership didn't matter, how big could this statistic get? (it is formed by disturbing the (real WT/KO structure). (significance)

The bootstrap answers the question : given the data I collected, how much would this statistic wobble with a slightly different sample? It is formed by keeping group identity and resampling within groups. (precision)

Repeat with different effect sizes

```
results table for different high response KO observations:
p_high    T_skew    p_skew    95% CI T_skew    T_tail    p_tail    95% CI T_tail
0.02      0.3585    0.0007 [ 0.1361, 0.5720]    0.0200    0.1483 [-0.0160, 0.0560]
0.05      0.5417    0.0000 [ 0.3304, 0.7470]    0.0220    0.1137 [-0.0080, 0.0620]
0.1       0.7436    0.0000 [ 0.5625, 0.9340]    0.1100    0.0000 [ 0.0700, 0.1400]
0.2       0.4846    0.0000 [ 0.3161, 0.6624]    0.1720    0.0000 [ 0.1400, 0.2080]
(base) keerthikesavan@MacBook-Air Exercises Solved Stats %
```

Discussion

Which statistic becomes significant first as p_high increases?

At p_high = 0.02, p_skew = 0.0007 is below the standard 0.05 cutoff, while p_tail = 0.1483 is not. T_skew picks up the asymmetry earlier, before there are enough high-responders. T_tail only becomes significant once p_high reaches 0.10.

Which statistic gives the more stable confidence interval?

T-tail is the more stable

What happens when the high-response subpopulation is very rare?

At p_high = 0.02, only 20 out of 1000 KO points are high-responders. T_tail's Con interval (-0.0160, 0.0560) crosses 0, so it cannot ignore differences

T_skew's CI (0.1361, 0.5720) does not cross zero even at this small of an effect size testing detecting the asymmetry even when the subpopulation is unique